

KICK-OFF & PROCESS

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01

How we test an idea at KaxaNuk, from a written claim to an honest verdict — and the tools you can use to do it yourself.

KAXANUK SHARING KNOWLEDGE.

We are an **investment research and technology development firm.**

THE PROCESS

Eight steps, from a written hypothesis to a verdict at the paper-trading gate.

THE INVESTMENT LAB

Six modules. Data Curator, Refinery and Analyzer, plus Backtest Engine, Portfolio Construction, and Attribution Analysis.

THE BOOTCAMP

Seven sessions, S00 to S06, where you run that process yourself, on your own laptop.

**YOU'LL LEAVE WITH
AN IDEA TESTED
CORRECTLY,
NOT NOTES.**

Every session ends with something in your repo. This one ends with
OBJECTIVE.md and the benchmark entry.

SEVEN SESSIONS.

S00 routes you through the pillars and sets up your stack.
S01 to S06 take one idea from a written claim to a verdict.

00 ONBOARDING & SELF-CHECK YOU LEAVE WITH A working stack, ROUTE.md, a researcher home.	01 KICK-OFF & PROCESS YOU LEAVE WITH A strategy repository, OBJECTIVE.md, the benchmark entry.	02 INVESTMENT RESEARCH YOU LEAVE WITH A five-part hypothesis, claims rewritten from notes.	03 FEATURE ENGINEERING YOU LEAVE WITH A point-in-time universe, one feature screened and counted.	04 PORTFOLIO CONSTRUCTION YOU LEAVE WITH BLUEPRINT_1.md pushed first, an equal-weight book that passes its invariants.	05 BACKTEST & ATTRIBUTION YOU LEAVE WITH Engine-priced runs, a Brinson-Fachler attribution, FINDINGS_1.md.	06 FINAL STRATEGY PREP & PRESENTATION YOU LEAVE WITH The challenge, N, a deflated Sharpe, the gate verdict, a rehearsed talk.
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THE RESEARCH PROCESS

Research is a pipeline, not a hunch.

EIGHT STEPS. ONE REPO.

Each step owns its outputs and reads only from the step above it. **Steps 1–7 live in your repo.** Step 8 is outside the course and the repo.

1**BIBLIOTHECA**

What do we believe, and on what evidence?

2**UNIVERSE**

Which securities are investable, point-in-time?

3**DATA**

Curation, refinery, analysis — what can we measure?

4**PORTFOLIO**

How is the book constructed?

5**BACKTEST**

How would it have performed?

6**ATTRIBUTION**

Where do the return and the risk come from?

7**PAPER TRADING**

Does it hold up on live data, before real money?

8**PRODUCTION**

Outside the course and outside the repo.

BIBLIOTHECA

Write down what you believe about the market, and the evidence behind it.

IN PLAIN WORDS

A literature review with a thesis at the end of it.

IT PRODUCES

A referenced hypothesis, in the repo, dated.

IT PREVENTS

Backtesting a hunch you cannot defend afterwards.



UNIVERSE

Decide which securities were investable, as of every date in the test.

IN PLAIN WORDS

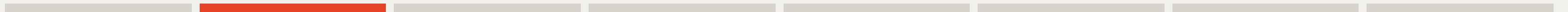
The eligible list, rebuilt for each date rather than today.

IT PRODUCES

A point-in-time membership table with liquidity filters.

IT PREVENTS

Survivorship bias — testing on the winners that survived.



DATA

Pull, clean and check the series you will actually compute on.

IN PLAIN WORDS

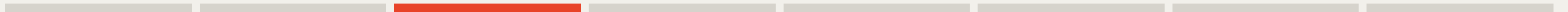
Curation, then refinery, then analysis — in that order.

IT PRODUCES

A reproducible dataset written to `Output/`.

IT PREVENTS

Beautiful results that came from broken inputs.



PORTFOLIO

Turn a signal into weights, limits and a rebalancing rule.

IN PLAIN WORDS

How much of what, and how often you change your mind.

IT PRODUCES

A weighting scheme with position and turnover limits.

IT PREVENTS

A good signal in a portfolio nobody could hold.



BACKTEST

Run the rules over history, with costs, without peeking ahead.

IN PLAIN WORDS

The simulation, run by the Backtest Engine.

IT PRODUCES

A performance curve and a cost-aware track record.

IT PREVENTS

Paper returns that real trading would have erased.



ATTRIBUTION

Split the return into known factor exposure and what is left.

IN PLAIN WORDS

Which part of the return did you actually earn?

IT PRODUCES

A factor and idiosyncratic breakdown of performance.

IT PREVENTS

Calling factor beta a return of your own.



PAPER TRADING

Run it forward on live data, with no money at risk.

IN PLAIN WORDS

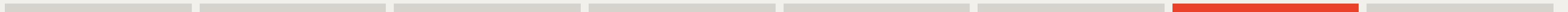
A dress rehearsal on data nobody has seen yet.

IT PRODUCES

Out-of-sample evidence and an operations checklist.

IT PREVENTS

Finding the plumbing problems once real money is at stake.



PRODUCTION.

The strategy leaves the repo, and this course.

IN PLAIN WORDS

Real capital, real monitoring, a real drawdown policy.

IT PRODUCES

A monitored strategy with an owner and a drawdown policy.

IT PREVENTS

Research that stays research forever.



THE TOOLS

Three tools, installed in S00. You use them in every session.

THREE JOBS, YOUR PICK.■

One tool per column. Either option does the job.

VERSIONS

Where your work lives, and how you get it back after you break it.

GITHUB DESKTOP

`github.com/apps/desktop`

GIT command line

`git-scm.com`

CODE & DEBUG

A command-capable AI assistant. It builds under a spec and tests you wrote first.

CLAUDE

`claude.ai/download`

CODEX

`openai.com/codex`

READ & RUN

Where you write and run Python. The free editions are enough.

PYCHARM

`jetbrains.com/pycharm`

VS CODE

`code.visualstudio.com`

ONE REPO, THREE JOBS.

The objective is to make it easy to build.

GITHUB

VERSIONS

Every state of your work, recoverable.

PYCHARM

READ & RUN

The editor, the interpreter and the terminal
in one window.

YOUR ASSISTANT

CODE & DEBUG

Explains errors and drafts code, under a
spec and tests you wrote first.

THE INVESTMENT LAB

Two ways in: the platform, or the library.

SIX MODULES.

One module per step of the process. Each reads the previous module's output, so you can enter the pipeline wherever your work already is.

01

DATA CURATOR

Pulls raw market and fundamental data from any provider and aligns it on one calendar.

YOUR S00 SMOKE TEST

02

DATA REFINERY

Cleans, adjusts and reshapes the curated data into analysis-ready series.

03

DATA ANALYZER

Builds features and tests whether they carry signal, before you model anything.

04

PORTFOLIO CONSTRUCTION

Turns a signal into weights, position limits and a rebalancing rule.

05

BACKTEST ENGINE

Runs the rules over history with costs and no look-ahead, and returns the track record.

06

ATTRIBUTION ANALYSIS

Splits the return into known factor exposure and the part that is actually yours.

NO CODE REQUIRED.

The same pipeline, driven from a workspace instead of a terminal: build a universe, download data, construct a portfolio, run a backtest and attribution — each step tracked to completion.

KaxaNuk
Sharing knowledge

Overview

UNIVERSE

UNIVERSE

- Creator
- Library

DATA

DATA

- Curator
- Analyzer

PORTFOLIO

PORTFOLIO

- Construction
- Experiments

BACKTEST

BACKTEST

- Parameters
- Experiments

ATTRIBUTION

ATTRIBUTION

- Attribution

SOON

Welcome back, Arturo

Here's your research workspace — pick up where you left off.

LICENSE

- Active

UNIVERSES

- 1

BACKTESTS

- 3

DATA FILES

- 26

Get started

Hide

Your pipeline is complete

4 of 4 completed

✓

Create a universe

Open

✓

Download market data

Open

✓

Build a portfolio

Open

✓

Run a backtest

Open

🕒

Review attribution

Understand what drove your returns.

SOON

CREATE IDEAS WITH CODE.

START HERE

`github.com/KaxaNuk`

```
uv add kaxanuk.data_curator
```

Open source, public on PyPI and GitHub — no account, no licence.

Two of the Lab's tools are open source and public: the Data Curator and the Researcher. An idea you can write as code is an idea you can run, share and test.

KaxaNuk-Researcher

Your researcher, the strategy template and a worked example. Set up in S00, used from today.

Data-Curator-Use-Cases

Worked examples on the free tool.

Backtest Engine · Attribution Analysis

On the 14-day Lab demo, or from reference outputs: sessions 04 to 06.

THE TOOLS IN USE

One worked example: from zero
to a clean price dataset.

WHAT THE DATA CURATOR DOES.

You describe the data you want — provider, tickers, dates, columns. It fetches, aligns and writes it, the same way every run.

01**YOU CONFIGURE**

Provider, tickers, dates and the columns you want.

02**IT CURATES**

Fetches from the provider, aligns the calendar, handles splits.

03**YOU GET A CSV**

In Output / , reproducible by anyone with your config.

LET THE ASSISTANT INSTALL IT.

PASTE THIS INTO YOUR ASSISTANT

I want to download market data using Yahoo Finance and KaxaNuk Data Curator, in a uv project on Python 3.13. Please help me install:
`https://kaxanuk-data-curator.readthedocs.io/en/latest/user_guide/quick_start.html`

It reads the quick-start, checks your Python version, runs the install and creates the project. You answer its questions and watch the terminal.

WHAT IT WILL RUN

```
$ uv init --python 3.13
$ uv add kaxanuk.data_curator
$ uv add
kaxanuk.data_curator_extensions.yahoo_finance
$ uv run kaxanuk.data_curator init excel
```

WHY YAHOO FINANCE

No API key. Thirty laptops, zero sign-ups.

INIT CREATES `Config/` · `Config/.env` · `Output/`

`Config/.env` is never committed and never printed.

TO DO.

01 READ ONE SOURCE PER CLAIM

Open access, into Bibliotheca/ with read. Part B of the order of work.

02 PUSH YOUR STRATEGY

OBJECTIVE.md and the benchmark entry in JOURNAL_1.md. Never Config/.env.

03 WRITE ONE BELIEF

One paragraph on something you think is true about markets, by hand. Session 02 turns it into a five-part hypothesis.

BACK TO THE PILLARS.

The coding, markets and math sessions behind today. Go back to one when you could not do its part with the AI off.

CODING · CODING FOUNDATIONS

- CF S01 Terminal, Environment and Git
git, a clean clone, uv and its lock file, .env never printed
- CF S04 The Agent Loop: Plan, Go, Review, Commit
the KaxaNuk Researcher: plan, go, review, commit

MARKETS · FINANCIAL MARKETS

- ▣ FM S01 The Ecosystem and the Security Master
the security master, dated and current sector
- ▣ FM S06 Indices, Benchmarks, ETFs and Survivorship
the benchmark, chosen first
- ▣ FM S12 Capstone: Performance, Attribution and Track Records
reading the worked example's CAGR and Sharpe

MATH · MATHEMATICAL FINANCE

- ▣ MF S01 Return Arithmetic
reading the worked example's CAGR and Sharpe

SEE YOU AT THE NEXT SESSION.

Bring your notes, the repo link
and the belief you wrote down.

QUESTIONS research@kaxanuk.mx

NEXT

SESSION 02

TOPIC

**INVESTMENT
RESEARCH: THE
HYPOTHESIS
WORKSHOP**

YOU LEAVE WITH

**A FIVE-PART
HYPOTHESIS, AND
CLAIMS REWRITTEN
FROM YOUR NOTES.**

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